

We also denote the induced pairing by $\langle \cdot, \cdot \rangle_A$, which satisfies

$$(2.24) \quad \langle m, \phi \rangle_A = \sum_{i=1}^k a_i (\phi(q_i) - \phi(p_i)), \quad m = \sum_{i=1}^k a_i (\delta_{q_i} - \delta_{p_i}) \in A_0(X, e), \quad \phi \in \text{Lip}_b(X),$$

understanding that $\phi(e) = 0$.

Proposition 2.11. Let $X \subset \mathbb{R}^n$ be a closed set and $e \notin X$. Then,

- (a) $A(X, e) \cong \mathcal{E}(X)$ via $m \mapsto m \llcorner X$ for $m \in A_0(X, e)$, extending by density,
- (b) $\text{Lip}_b(X) \simeq \mathcal{E}(X)^*$ via $\phi \mapsto (m \mapsto \langle m, \phi \rangle_{\mathcal{E}})$.

Proof. By [Wea18, Thm. 3.3, Cor. 3.4], the pairing (2.24) extends to an isometric isomorphism $A(X, e)^* \cong \text{Lip}_b(X)$. In particular this implies that for $m \in A_0(X, e)$,

$$(2.25) \quad \|m \llcorner X\|_{\text{Lip}_b(X)^*} = \sup\{|\langle m, \phi \rangle_A| : \|\phi\|_{\text{Lip}_b(X)} \leq 1\} = \|m\|_A.$$

Hence it follows that $A(X, e) \cong \mathcal{E}(X)$ by sending each $m \in A_0(X, e)$ to $m \llcorner X \in \mathcal{E}_0(X)$ and extending by density, thereby proving (a). Since we also have $\langle m, \phi \rangle_A = \langle m \llcorner X, \phi \rangle_{\mathcal{E}}$ for all $m \in A_0(X, e)$ and $\phi \in \text{Lip}_b(X)$, it follows that $\mathcal{E}(X)^* \cong \text{Lip}_b(X)$ via the pairing $\langle \cdot, \cdot \rangle_{\mathcal{E}}$, establishing (b). $\square$

Example 2.12. If $X \subset \mathbb{R}^n$ is closed, we have $\mathcal{M}(X) \subset \mathcal{E}(X)$, by noting for that each $\mu \in \mathcal{M}(X)$, the mapping $\phi \mapsto \int_X \phi d\mu$ is well-defined and weakly*-continuous on $\text{Lip}_b(X)$. However this space is strictly larger in general; if $a \in X$ is an accumulation point of X , then we can find a sequence $(a_k)_k \subset X$ converging to a such that $a_k \neq a$ for all k . By passing to a subsequence if necessary, assume that $\sum_k |a_k - a| < \infty$. Then $m = \sum_{k=1}^{\infty} (\delta_{a_k} - \delta_a) \in \mathcal{E}(X)$ by noting the series converges absolutely in $\text{Lip}_b(X)^*$.

Lemma 2.13. Let $X \subset \mathbb{R}^n$ be any set, and $\phi_k, \phi \in \text{Lip}_b(X)$. Then, as $k \rightarrow \infty$,

$$(2.26) \quad \phi_k \xrightarrow{*} \phi \text{ in } \text{Lip}_b(X) \iff \begin{cases} \phi_k \rightarrow \phi \text{ uniformly on bounded subsets of } X, \\ \sup_k \|\phi_k\|_{\text{Lip}_b(X)} < \infty. \end{cases}$$

Proof. Using the identification $\text{Lip}_b(X) \cong \text{Lip}_b(\overline{X})$, we can assume without loss of generality that X is closed. If $\phi_k \xrightarrow{*} \phi$ weakly* in $\text{Lip}_b(X)$, by the Banach-Steinhaus theorem, we have $\|\phi_k\|_{\text{Lip}_b(X)}$ is uniformly bounded in k . Then by applying the Arzelà-Ascoli theorem, there is a subsequence ϕ_{k_j} which converges uniformly to ϕ on $X \cap B_M(0)$ for each $M \in \mathbb{N}$, and hence $\phi_{k_j} \rightarrow \phi$ uniformly on bounded subsets of X . Since the limit is unique, this convergence also holds for the entire sequence ϕ_k .

Conversely since $\text{Lip}_b(X)$ is the dual of a separable space, the weak*-topology is compact and metrisable on norm-bounded subsets (see e.g. [Bre11, Thm. 3.16, 3.28]). Therefore ϕ_k admits a weakly*-convergent subsequence, but since this limit is uniquely determined as ϕ using the uniform convergence, the entire sequence ϕ_k converges weakly* to ϕ . $\square$

We will often use Lemma 2.13 with $X = [0, 1]$, noting that $\text{Lip}_b([0, 1]) = W^{1,\infty}((0, 1))$. For general open sets U however, we have a slightly different characterisation for weak* convergence in $W^{1,\infty}(U)$.

Lemma 2.14. Let $U \subset \mathbb{R}^n$ be an open set. Then if $\phi_k, \phi \in W^{1,\infty}(U)$, as $k \rightarrow \infty$,

$$(2.27) \quad \phi_k \xrightarrow{*} \phi \text{ in } W^{1,\infty}(U) \iff \begin{cases} \phi_k \rightarrow \phi \text{ pointwise,} \\ \sup_k \|\phi_k\|_{W^{1,\infty}(U)} < \infty. \end{cases}$$

In addition, the space $C_b^1(U)$ is sequentially weakly* dense in $W^{1,\infty}(U)$.

We note that $C_b^1(U) \not\subset \text{Lip}_b(U)$ in general, so we do not get an analogous density statement there. Also the below proof shows in fact that $\phi_k \rightarrow \phi$ locally uniformly in U in (2.27), however pointwise convergence will suffice for our purposes.